

Data Pipeline Exercise

Evidence of Execution — Olist E-Commerce ELT Pipeline

*Azure Data Factory + Self-Hosted Integration Runtime + Azure Databricks + Azure SQL
Database*

Cloud run: 2026-07-30

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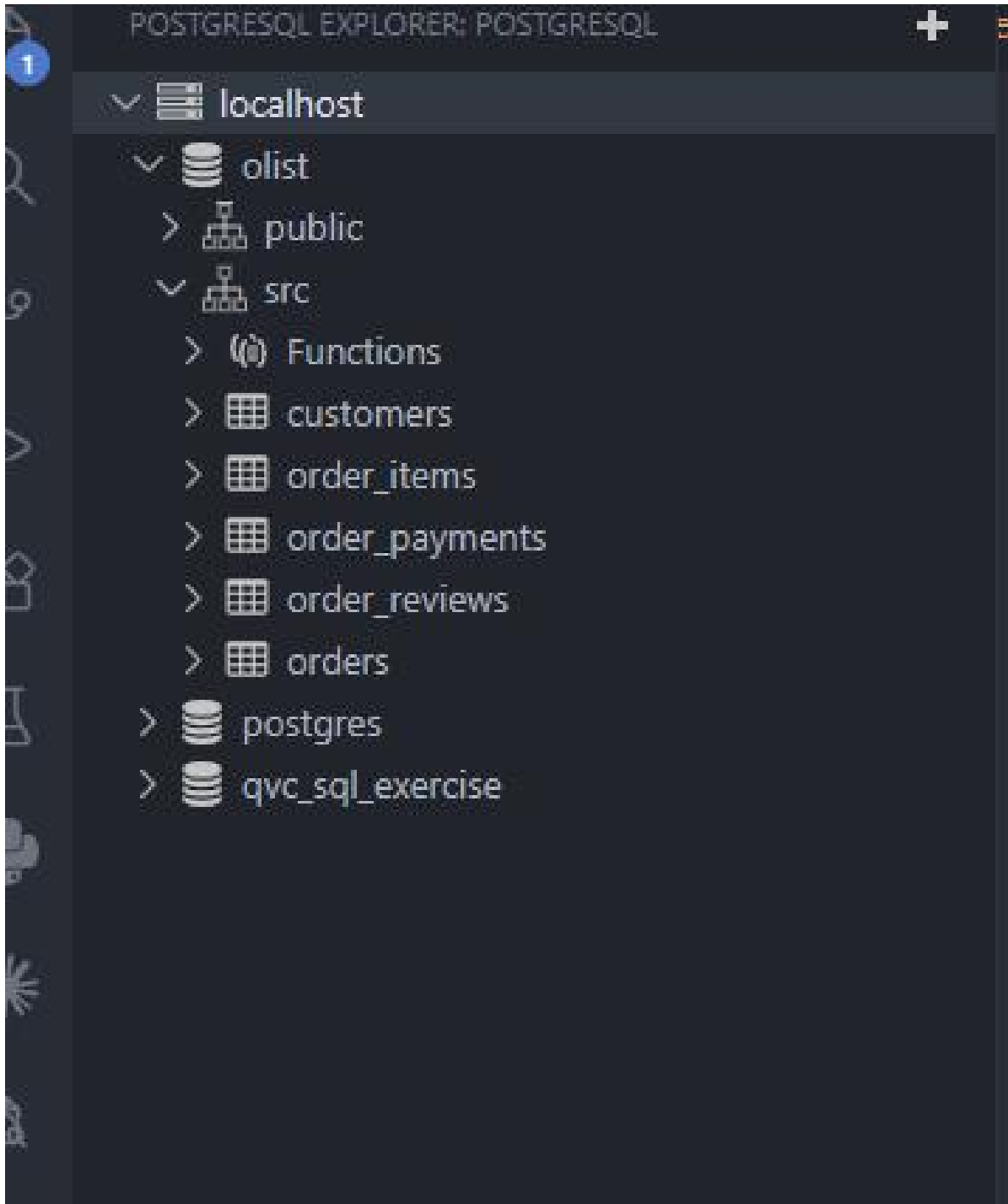
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1. Source Systems

Confirms the relational source (Docker PostgreSQL) exists and is seeded with the exact expected row counts before the pipeline runs.

PostgreSQL 16 (Docker) — olist database, src schema, 5 transactional tables.



Row counts confirmed at source: orders 99,441 · customers 99,441 · order_items 112,650 · order_payments 103,886 · order_reviews 99,224.

med-query-1 SELECT 'orders' AS t, COUNT(*) FROM src. Untitled-2 Results: untitled:untitled-2

```
1 SELECT 'orders' AS t, COUNT(*) FROM src.orders
2 UNION ALL SELECT 'customers', COUNT(*) FROM src.customers
3 UNION ALL SELECT 'order_items', COUNT(*) FROM src.order_items
4 UNION ALL SELECT 'order_payments', COUNT(*) FROM src.order_payments
5 UNION ALL SELECT 'order_reviews', COUNT(*) FROM src.order_reviews
```

5 rows returned

	t text	count bigint
1	order_payments	103886
2	customers	99441
3	order_reviews	99224
4	order_items	112650
5	orders	99441

2. ADF Linked Services

Both linked services authenticate with secrets typed directly in — no Key Vault, no role assignments (see README Assumptions).

PostgreSQL linked service via self-hosted-shir — connection successful.

New linked service

 PostgreSQL [Learn more](#) 

Name *

Description

Connect via integration runtime * 

 self-hosted-shir



 The credentials are stored in the machines of self-hosted integration runtime if you don't choose to store them in Azure Key Vault.

Server name *

Port

Database name *

User name *

Password

Azure Key Vault

Create

Back

 Connection successful

 Test connection

Cancel

Azure Databricks linked service via Access Token — connection successful.

e



1

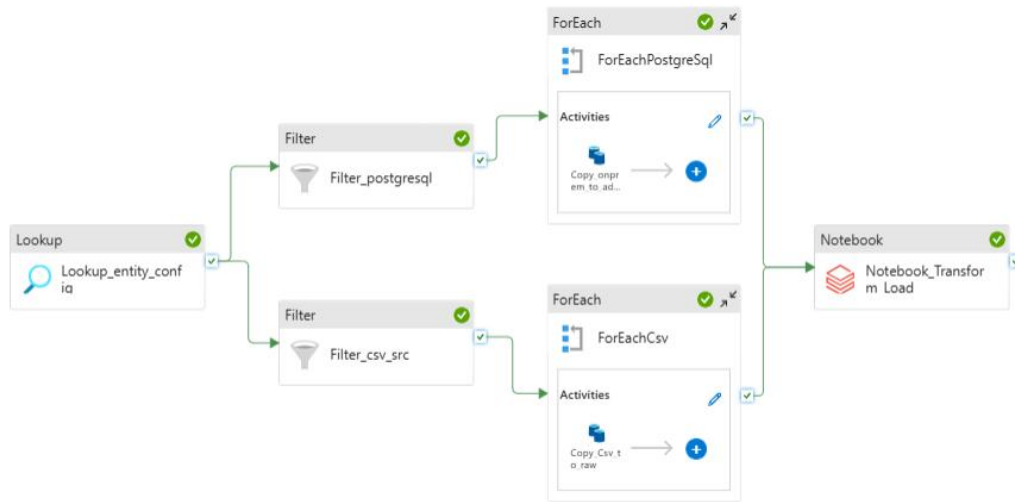
[illegible]

100

3. ADF Pipeline — Design & Execution

The metadata-driven pipeline: one config file (entities.json) drives 9 parameterised Copy activities across two source types.

pLolist_ingest: Lookup → Filter ×2 → ForEach ×2 (9 parameterised copies) → Notebook_Transform_Load — all activities succeeded.



4. ADLS Raw Zone

Every run lands under a RunId-scoped path — dumb ingestion, full fidelity retained before any transformation.

raw/ — csv and parquet landing folders.

Blob containers

>

olistdata

>

raw

Authentication method: Access key (Switch to Microsoft Entra user account)

Search blobs by prefix (case-sensitive)

Showing all 3 items

Name

[.]

csv

parquet

raw/parquet/ — run-scoped Postgres extracts, named <entity><RunId>.

datasource4dbs

Favorites

Recently viewed

Blob containers

\$blobchangefeed

\$logs

employee

inbox

olistdata

View all

Classic file shares

Queues

Tables

+ Add Directory

↑ Upload

↻ Refresh

🗑 Delete

📄 Copy

📄 Paste

🏷 Rename

🔑 Acquire lease

🔑 Break lease

⋮

Blob containers

>

olistdata

>

raw

>

parquet

Authentication method: Access key (Switch to Microsoft Entra user account)

Search blobs by prefix (case-sensitive)

Only show active blobs

Showing all 5 items

	Name	Last modified	Access tier	Blob type
	[.]			
	customers083ead2f-75c9-4bf8-8a76-80da2c1bf691	7/30/2026, 8:37:56 AM	Hot (Inferred)	Block blob
	order_items083ead2f-75c9-4bf8-8a76-80da2c1bf691	7/30/2026, 8:38:02 AM	Hot (Inferred)	Block blob
	order_payments083ead2f-75c9-4bf8-8a76-80da2c1bf691	7/30/2026, 8:37:52 AM	Hot (Inferred)	Block blob
	order_reviews083ead2f-75c9-4bf8-8a76-80da2c1bf691	7/30/2026, 8:38:02 AM	Hot (Inferred)	Block blob
	orders083ead2f-75c9-4bf8-8a76-80da2c1bf691	7/30/2026, 8:38:02 AM	Hot (Inferred)	Block blob

5. Azure SQL — Target Schema (DDL)

Confirms sql/01_azure_sql_ddl.sql ran successfully: curated/etl/mart schemas, 9 curated tables, 3 mart views.

curated schema — all 9 typed tables created.

DB_SUPPLY_CHAIN (bhanutejachindukuri@gmail.com)

▼ Tables

> curated.customers

> curated.geolocation

> curated.order_items

> curated.order_payments

> curated.order_reviews

> curated.orders

> curated.product_category_translation

> curated.products

> curated.sellers

mart schema — all 3 consumption views created.

> mart.v_payment_mix

> mart.v_review_delivery

> mart.v_sales_overview

> sys.database_firewall_rules

...

...

...

...

6. Databricks Transform Run

The PySpark transform layer: rename/cast/dedup/DQ-flag logic, run against the full real dataset.

transform_olist notebook job — Succeeded in 6m 2s; per-entity read/written/flagged counts.

The screenshot shows the Databricks interface for a job run. At the top, there are buttons for "Delete job run" and "Repair run". The main area is divided into "Output" and "Details".

Output: The output shows a PySpark code snippet and its execution results. The code snippet is:

```
n_processed += 1
print(f"[OK] {entity:<30} read={n_read:>9,} written={n_written:>9,} flagged={n_flagged}>
")
except Exception as exc:
    failures.append(entity)
    log_process(entity, "curated_write", None, None, None,
                "FAILED", str(exc)[:900], t0, datetime.now(timezone.utc))
    print(f"[FAILED] {entity}: {exc}")
```

The execution results show a table of per-entity read/written/flagged counts:

Entity	read	written	flagged
orders	99,441	99,441	0
customers	99,441	99,441	0
order_items	112,650	112,650	0
order_payments	103,886	103,886	3
order_reviews	99,224	99,224	0
products	32,951	32,951	0
sellers	3,095	3,095	0
product_category_translation	71	71	0

Details: The details section shows the job ID (678303358625913), task run ID (86005145625200), run as (bhanutejachindukuri), started (Jul 30, 2026, 10:33 AM), ended (Jul 30, 2026, 10:39 AM), duration (6m 2s), and status (Succeeded). There are buttons for "View run events" and "View run libraries".

Job run detail — job ID, task run ID, notebook path, status Succeeded.

The screenshot shows the Databricks interface for a job run. At the top, there are buttons for "Delete job run" and "Repair run". The main area is divided into "Output" and "Details".

Output: The output shows a notebook titled "Olist transform notebook (raw → curated)". The notebook content is:

```
Reads the raw files landed by ADF in ADLS — csv reference files under <raw base>/csv/<entity>/,
Postgres-sourced parquet flat under <raw base>/parquet/ named <entity><RunId> (see read_raw below
for why the two shapes differ) — applies the transformations defined in olist_transforms (rename / cast
/ null handling / dedup / DQ flags / ingestion metadata), writes typed tables to Azure SQL via JDBC, and
logs per-entity row counts to etl.pipeline_process_log.

Parameters are passed by the ADF Notebook activity via widgets. Import BOTH files into the same
workspace folder (e.g. /Shared/ ): this notebook %run s ./olist_transforms.
```

The execution results show a table of per-entity read/written/flagged counts:

Entity	read	written	flagged
orders	99,441	99,441	0
customers	99,441	99,441	0
order_items	112,650	112,650	0
order_payments	103,886	103,886	3
order_reviews	99,224	99,224	0
products	32,951	32,951	0
sellers	3,095	3,095	0
product_category_translation	71	71	0

Details: The details section shows the job ID (678303358625913), task run ID (86005145625200), run as (bhanutejachindukuri), started (Jul 30, 2026, 10:33 AM), ended (Jul 30, 2026, 10:39 AM), duration (6m 2s), and status (Succeeded). There are buttons for "View run events" and "View run libraries".

7. ADF Monitor — Pipeline & Activity Runs

Monitor tab evidence of the pipeline executing end-to-end in the cloud, not just in Debug.

Activity-level run detail — every Copy/ForEach/Notebook activity succeeded.

All pipeline runs > pl_ol_ingest_onprem_to_adls - Activity runs

[Refresh](#) [Update pipeline](#) [List](#) [Gantt](#)

! Pipeline was modified after this run. The current pipeline configuration is shown.

Activity name	Activity st...	Activit...	Run start	Duration	Integration runtime	User prop...	Activity run ID
Notebook_Transform_Load	✓ Succeeded	Notebook	7/30/2026, 10:33:39 AM	6m 42s	AutoResolveIntegrationRuntime (East US)		726f1555-ac99
Copy_onprem_to_adls_raw	✓ Succeeded	Copy data	7/30/2026, 10:32:37 AM	54s	self-hosted-shir		cfa3e35e-81c6
Copy_onprem_to_adls_raw	✓ Succeeded	Copy data	7/30/2026, 10:32:37 AM	58s	self-hosted-shir		8e3708e8-078c
Copy_onprem_to_adls_raw	✓ Succeeded	Copy data	7/30/2026, 10:32:37 AM	50s	self-hosted-shir		dffff2af-381e-4
Copy_onprem_to_adls_raw	✓ Succeeded	Copy data	7/30/2026, 10:32:37 AM	58s	self-hosted-shir		1dbc3bfe-abad
Copy_onprem_to_adls_raw	✓ Succeeded	Copy data	7/30/2026, 10:32:37 AM	59s	self-hosted-shir		329ea158-a364
Copy_Csv_to_raw	✓ Succeeded	Copy data	7/30/2026, 10:32:36 AM	19s	AutoResolveIntegrationRuntime (East US)		807f5cd5-f412
Copy_Csv_to_raw	✓ Succeeded	Copy data	7/30/2026, 10:32:36 AM	18s	AutoResolveIntegrationRuntime (East US)		4f424080-aa61
Copy_Csv_to_raw	✓ Succeeded	Copy data	7/30/2026, 10:32:36 AM	16s	AutoResolveIntegrationRuntime (East US)		081af410-7ae9
Copy_Csv_to_raw	✓ Succeeded	Copy data	7/30/2026, 10:32:36 AM	20s	AutoResolveIntegrationRuntime (East US)		82acf1b5-5466
ForEachPostgreSql	✓ Succeeded	ForEach	7/30/2026, 10:32:36 AM	1m 3s			48b5b52d-eed1
ForEachCsv	✓ Succeeded	ForEach	7/30/2026, 10:32:34 AM	24s			9c80d5b0-5681
Filter_postgresql	✓ Succeeded	Filter	7/30/2026, 10:32:33 AM	2s			58f47c69-c14e
Filter_csv_src	✓ Succeeded	Filter	7/30/2026, 10:32:33 AM	Less than 1s			87d4a0e2-33f2

Pipeline run history — 12 runs shown, including earlier debug iterations.

Pipeline runs

Triggered **Debug** [Rerun](#) [Cancel options](#) [Refresh](#) [Edit columns](#) [List](#) [Gantt](#)

Filter by run ID or name Local time: **Last 7 days** Pipeline name: **All** Status: **All** [Add filter](#) [Copy filters](#) [Export to CSV](#)

Showing 1 - 12 items Last refreshed 0 minutes ago

☐	Pipeline name	Run start	Run end	Duration	Status	Triggered by	Run ID	Parameters
☐	pl_ol_ingest_onprem_t...	7/30/2026, 10:32:14	7/30/2026, 10:40:22	8m 9s	✓ Succeeded	Manual trigger	57fb0a03-e3b2-44d3-...	
☐	pl_ol_ingest_onprem_t...	7/30/2026, 10:20:35	7/30/2026, 10:22:30	1m 55s	✓ Succeeded	Manual trigger	b072ba6c-7cb1-4a89-...	
☐	pl_ol_ingest_onprem_t...	7/30/2026, 10:18:59	7/30/2026, 10:19:24	25s	⊗ Canceled	Manual trigger	2f3e49bf-eb63-42b8-8...	
☐	pl_ol_ingest_onprem_t...	7/30/2026, 8:37:00 A	7/30/2026, 8:38:10 A	1m 11s	✓ Succeeded	Manual trigger	083ead2f-75c9-4bf8-8...	
☐	pl_ol_ingest_onprem_t...	7/30/2026, 8:31:40 A	7/30/2026, 8:34:17 A	2m 37s	✓ Succeeded	Manual trigger	9814a072-b02d-44df-...	
☐	pl_ol_ingest_onprem_t...	7/30/2026, 8:26:38 A	7/30/2026, 8:29:11 A	2m 34s	✗ Failed	Manual trigger	319e6c4c-62dd-459b-...	
☐	pl_ol_ingest_onprem_t...	7/30/2026, 8:14:29 A	7/30/2026, 8:14:49 A	20s	✓ Succeeded	Manual trigger	2e075a9f-18a4-432f-8...	
☐	pl_ol_ingest_onprem_t...	7/30/2026, 8:10:41 A	7/30/2026, 8:11:11 A	30s	✓ Succeeded	Manual trigger	9833d592-2242-409a-...	
☐	pl_ol_ingest_onprem_t...	7/30/2026, 8:09:08 A	7/30/2026, 8:09:08 A	Less than 1s	✗ Failed	Manual trigger	9a411228-42e1-43de-...	
☐	pl_ol_ingest_onprem_t...	7/30/2026, 8:08:57 A	7/30/2026, 8:08:58 A	Less than 1s	✗ Failed	Manual trigger	172ee824-6db2-4019-...	
☐	pl_ol_ingest_onprem_t...	7/30/2026, 8:06:00 A	7/30/2026, 8:06:20 A	21s	✓ Succeeded	Manual trigger	4000b545-43c7-4c43-...	
☐	pl_ol_ingest_onprem_t...	7/30/2026, 8:05:20 A	7/30/2026, 8:05:43 A	23s	✓ Succeeded	Manual trigger	af403048-8103-4c8a-b...	

8. Process Log Reconciliation

Per-entity reconciliation written by the notebook on every run — the mechanism behind every row count cited in the README.

etl.pipeline_process_log for the latest run — all 9 entities, rows_read/rows_written/rows_flagged, status SUCCESS.

Query 1 ×

▶ Run

☐ Cancel query

⬇ Save query

⬇ Export data as

▼

⌘ Show only Editor

1

2

3

4

5

6

7

SELECT

entity_name,

stage,

rows_read,

rows_written,

rows_flagged,

status,

started_at_utc,

finished_at_utc

FROM

etl.pipeline_process_log

WHERE

pipeline_run_id =

(SELECT TOP 1 pipeline_run_id

FROM etl.pipeline_process_log

ORDER BY logged_at_utc DESC)

ORDER BY

logged_at_utc;

Results

Messages

entity_name	stage	rows_read	rows_written	rows_flagged	status	started_at_utc	finished_at_utc
orders	curated_write	99441	99441	0	SUCCESS	2026-07-30T08:33:52....	2026-07-30T08:34:54.
customers	curated_write	99441	99441	0	SUCCESS	2026-07-30T08:35:06....	2026-07-30T08:35:34.
order_items	curated_write	112650	112650	0	SUCCESS	2026-07-30T08:35:42....	2026-07-30T08:36:11.
order_payments	curated_write	103886	103886	3	SUCCESS	2026-07-30T08:36:19....	2026-07-30T08:36:48.

9. Data Quality Distributions

Direct proof of the 'flag, don't drop' design principle — every number here is queried live against the curated tables.

1/4 — *orders.is_valid_delivery_date*: 99,441 valid, 0 flagged.

[Login](#) [New Query](#) [Open query](#) [Open in](#) [Feedback](#) [Getting started](#) [Modern experience](#)

>>

Query 1 ×

[Run](#) ☐ Cancel query [Save query](#) [Export data as](#) [Show only Editor](#)

```
1 SELECT is_valid_delivery_date, COUNT(*) AS n
2 FROM curated.orders
3 GROUP BY is_valid_delivery_date;
```

Results

Messages

is_valid_delivery_date	n
True	99441

2/4 — *order_payments* by type: exactly 3 'not_defined' rows, kept and flagged, not dropped.

>>

Query 1 ✕

Run Cancel query Save query Export data as Show only Editor

```
1 SELECT payment_type, is_valid_payment, COUNT(*) AS n
2 FROM curated.order_payments
3 GROUP BY payment_type, is_valid_payment
4 ORDER BY n DESC;
```

Results Messages

Search to filter items...

payment_type	is_valid_payment	n
credit_card	True	76795
boleto	True	19784
voucher	True	5775
debit_card	True	1529
not_defined	False	3

3/4 — order_reviews score distribution (1..5), all valid.

<< Login + New Query ↑ Open query Open in Feedback Getting started Modern experience

>>

Query 1 ✕

Run Cancel query Save query Export data as Show only Editor

```
1 SELECT review_score, is_valid_review_score, COUNT(*) AS n
2 FROM curated.order_reviews
3 GROUP BY review_score, is_valid_review_score
4 ORDER BY review_score;
```

Results Messages

Search to filter items...

review_score	is_valid_review_score	n
1	True	11424
2	True	3151
3	True	8179
4	True	19142
5	True	57328

4/4 — geolocation compression proof: 19,015 zip prefixes, 1,000,163 raw points, 29 excluded.



Query 1 ×

[Run](#) ☐ [Cancel query](#) [Save query](#) [Export data as](#) [Show only Editor](#)

```
1 SELECT COUNT(*) AS zip_prefixes,  
2     SUM(point_count) AS raw_points,  
3     SUM(invalid_point_count) AS excluded_points  
4 FROM curated.geolocation;  
5
```

[Results](#) [Messages](#)

Search to filter items...

zip_prefixes	raw_points	excluded_points
19015	1000163	29

10. Consumption Layer — Mart Views

Proof that both source systems (relational Postgres + file-based CSVs) come together as one integrated, analytically-queryable dataset.

mart.v_sales_overview — top states × categories by order count.

Run Save as view

⚠ Your queries will be lost when your session ends. Use **Save as view** to keep results as a database view, or copy your query text to

```
1 SELECT TOP 10 customer_state, product_category,
2   order_count, gross_merchandise_value, avg_delivery_days
3 FROM mart.v_sales_overview
4 ORDER BY order_count DESC;
```

Messages Results

📄 {} ⬇

📄 🔍 ⬆

	ABC	customer_state	ABC	product_category	123	order_count	e*	gross_merchandise_va...	e*	avg_delivery_days
1		SP		bed_bath_table		4347		472238.07		9.152026
2		SP		health_beauty		3715		453916.48		8.146666
3		SP		sports_leisure		3216		375044.87		8.624440
4		SP		housewares		2720		269656.60		7.916143

mart.v_payment_mix — full 4-row output.

⚠ Your queries will be lost when your session ends. Use **Save as view** to keep results as a database view, or copy your query text to

```
1 SELECT * FROM mart.v_payment_mix ORDER BY total_payment_value DESC;
```

Messages Results

📄 {} ⬇

📄 🔍 ⬆

	ABC	payment_type	123	order_count	123	payment_count	e*	total_payment_v...	e*	avg_installments	e*	pct_of_value
1		credit_card		76505		76795		12542084.19		3.507155		78.34
2		boleto		19784		19784		2869361.27		1.000000		17.92
3		voucher		3866		5775		379436.87		1.000000		2.37
4		debit_card		1528		1529		217989.79		1.000000		1.36

mart.v_review_delivery — full 5-row output.

Run Save as view

⚠ Your queries will be lost when your session ends. Use **Save as view** to keep results as a database view, or copy your query text to

```
1 SELECT * FROM mart.v_review_delivery ORDER BY review_score;
```

Messages

Results



	123 review_score	123 review_count	e ^x avg_delivery_days	e ^x pct_delivered_late
1	1	9406	21.251887	37.78
2	2	2941	16.605916	20.61
3	3	7961	14.202361	10.99
4	4	18987	12.253120	5.02
5	5	57066	10.625422	3.00

11. Schedule Trigger

Demonstrates scheduling capability; stopped so it won't fire unattended (tumbling-window trigger would be the production choice — see README Productionising).

Daily_schedule_olist_trigger — *ScheduleTrigger*, weekly recurrence on Mon/Fri, stopped.

The screenshot shows the 'Edit trigger' configuration page in the Microsoft Fabric Data Factory portal. The left sidebar displays the 'Triggers' section with a table listing the trigger 'Daily_schedule_olist_trigger' of type 'Schedule'. The main panel shows the configuration details for this trigger.

Edit trigger

Name *
Daily_schedule_olist_trigger

Description

Type *
ScheduleTrigger

Start date *
7/31/2026, 11:46:00 AM

Time zone *
Amsterdam, Berlin, Bern, Rome, Stockholm, Vienna (UTC+2)

Recurrence *
Every 1 Week(s)

Advanced recurrence options

Run on these days
Sun Mon Tue Wed Thu Fri Sat

Execute at these times

OK **Cancel**